

Skills Governance Audit

Consolidation Plan & Gap Analysis

1. Duplication Matrix

Y=present, N=absent. Recommended sole owner shown in Section 5.

Rule	arch	domain	safe	test	state
Project mission (not a screener)	Y	Y	N	N	Y
Transactions = source of truth	Y	Y	N	N	Y
Never bypass transaction history	Y	Y	N	N	Y
Portfolio data flow diagram	Y	N	N	N	Y
Holdings are derived	Y	Y	N	N	Y
Allocation = reporting layer only	N	Y	N	N	Y
Portfolio accounting transaction types	Y	N	N	N	Y
Currency: convert first then aggregate	Y	Y	N	N	Y
Supported currencies (SAR/USD/AED/EUR)	Y	Y	N	N	N
Market data must go through layers	Y	N	N	N	Y
Saudi routing (prefer SAHMK, fallback Yahoo)	Y	N	N	N	N
Research modules are read-only	Y	Y	N	N	N
MVP tier priorities	Y	N	N	N	Y
Safe zones component list	Y brief	N	Y full	N	Y brief
Review First Mode (full checklist)	Y full	N	N	N	Y partial
Change management checkpoint format	Y	N	N	N	N
Test count must never decrease	Y	N	N	Y	Y
Test categories list	Y(10)	N	N	Y(11)	N
Filter rule (filters != accounting)	N	Y	Y	N	N
Risk classification P0-P4	N	N	Y	N	N
Escalation rule	N	N	Y	N	N
Minimal change principle	N	N	Y	N	N
Mobile First	Y	N	N	N	N
Roles (PM/Architect/Engineer)	Y	N	N	N	N
Mandatory loading sequence	N	N	N	N	Y
Test runner registration rule	N	N	N	Y	N
Failure / completion rule	N	N	N	Y	N

Duplication summary by skill:

Skill	Duplicated elsewhere	Unique rules
bousala-architecture	14 rules duplicated	5 unique
bousala-investment-domain	8 rules duplicated	3 unique
bousala-safe-zones	2 rules duplicated	4 unique
bousala-test-protocol	2 rules duplicated	5 unique
bousala-project-state	13 rules duplicated	3 unique

2. Conflict Matrix

ID	Severity	Description
C1	Medium	CASH test category in test-protocol but absent from architecture. Architecture lists 10 categories; test-protocol lists 11 (includes CASH). Cash is Tier 1 critical.
C2	Medium	SAHMK described as 'Planned' in project-state but treated as existing protected zone in architecture and safe-zones. SAHMK is already implemented and tested.

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C3	Low	Review First output fields differ. architecture omits 'recommendation' and 'architecture impact'; project-state includes both. Different sessions produce different review formats.
C4	Low	Approval trigger phrase 'Proceed with implementation' defined in architecture but no skill clarifies if shorter phrases (ok, yes, go) are valid. Caused ambiguity in practice.
C5	Low-Med	Filter rule: safe-zones says 'visibility/reporting/chart/table scope'; investment-domain says 'UI and reporting only'. Neither explicitly covers session state keys -- root cause of the alloc_ms_market multi-session conflict.

3. Governance Gap Analysis

Gaps ranked by likelihood of causing a real future defect.

ID	Likelihood	Impact	Description
G1	High	High	Session state ownership rule absent. No skill prevents two UI controls sharing one session state key. The alloc_ms_market conflict required 3 sessions to resolve and WILL recur for any new filter pair.
G2	Med	High	dev_test_runner.py not listed as a protected zone. Removing or weakening existing tests has no escalation path. Test governance can be silently broken.
G3	Med	High	No portfolio data schema change governance. No backward-compat rule or migration requirement when holdings.json or any data file gains, loses, or renames a field.
G4	Med	Med	No post-implementation app validation checklist. Source-inspection tests can pass while the running app is broken. No rule requires: restart workflow, open preview, verify KPI values consistent.
G5	Med	Med	No breaking change definition. Removing a function, changing a calculation output, altering exported data structure -- none have a classification or explicit approval path.
G6	Low-Med	Med	Approval phrase ambiguity. No skill defines whether 'ok', 'yes', 'approved', or 'go' are valid triggers. 'Proceed with implementation' is the only named phrase.
G7	Low	Med	No rollback governance. Skills say create checkpoint but give no rule for when rollback is appropriate or who approves it.
G8	Low	Low	Mobile First rule buried inside architecture with no prominent placement. Easily skipped when scanning context.

4. Context Efficiency Analysis

Total lines currently loaded per full session: 1,077. Duplicate content: ~380 lines (35%). Recommended ceiling after consolidation: <=700 lines.

Skill	Lines	Load freq	Risk red.	Cost	Maint.	Classification
bousala-architecture	387	Every session	High	High	High	Valuable -- overloaded
bousala-investment-domain	161	Domain Qs	High	Low	Low	Essential -- high ROI
bousala-safe-zones	155	Every impl.	High	Low	Low	Essential -- highest ROI
bousala-test-protocol	136	Every impl.	High	Low	Low	Essential -- high ROI
bousala-project-state	238	Every session	Med	High	High	Valuable -- needs trimming

Balance assessment:

- Are skills increasing token consumption? YES -- 1,077 lines is 35% above the recommended 700-line ceiling.
- Is current skill count appropriate? YES -- five skills at the right scope is correct. Problem is content overlap, not count.
- Recommended max skills at current MVP stage: 5 skills, <=150 lines each, <=700 lines total.
- Highest ROI skill: bousala-safe-zones. Unique content (P0-P4, escalation), low cost, loaded every session.
- Lowest ROI skill: bousala-project-state in current form -- 80% is restatement of other skills.
- What stays in skills: rules that change agent BEHAVIOUR (Review First, safe zones, test protocol, domain model).
- What moves to documentation: project history, feature roadmap, changelog, implementation notes.

5. Recommended Ownership Map

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Target state: each rule lives in exactly one skill.

Rule class	Sole owner
Project mission, product vision	project-state
Mandatory loading sequence	project-state
Current implementation status (SAHMK, modules built)	project-state
Development stage and roadmap priorities	project-state
Long-term goal	project-state
Review First Mode (full checklist + approval phrase)	architecture
Change management format (checkpoint, post-impl report)	architecture
Core architecture diagram	architecture
Valuation single-source rule	architecture
Market data layer rule (no direct UI calls)	architecture
Saudi routing (SAHMK preferred, Yahoo fallback)	architecture
Research read-only rule	architecture
MVP tier priorities	architecture
Mobile First	architecture
Roles (PM/Architect/Engineer)	architecture
Transactions = source of truth	investment-domain
Holdings are derived + how	investment-domain
Allocation is reporting layer only	investment-domain
Cash rules (first-class asset, movements)	investment-domain
P&L rules (realized vs. unrealized, never mix)	investment-domain
Multi-currency rules	investment-domain
Accounts (multi-broker/bank)	investment-domain
Filter rule (filters do not change accounting)	investment-domain
Protected zone component list	safe-zones
Risk classification P0-P4	safe-zones
Escalation rule	safe-zones
Minimal change principle	safe-zones
UI / golden rule	safe-zones
Session state ownership rule [NEW -- closes G1]	safe-zones
dev_test_runner.py as protected zone [NEW -- closes G2]	safe-zones
Test count invariant	test-protocol
Test categories (all 11 including CASH)	test-protocol
Test runner registration rule	test-protocol
Completion and failure rules	test-protocol
Mandatory post-impl reporting format	test-protocol
Post-impl app validation checklist [NEW -- closes G4]	test-protocol

6. Exact Sections Recommended for Future Removal / Consolidation

From bousala-architecture (remove 4 sections, --80 lines):

- Remove 'Single Source Of Truth' -- sole ownership moves to investment-domain
- Remove 'Safe Zones' (brief list) -- already owned in full by safe-zones; brief copy adds drift risk
- Remove 'Testing Rules' -- already owned in full by test-protocol (fix CASH omission there first)
- Remove 'Currency Rules' -- sole ownership moves to investment-domain

From bousala-project-state (remove 9 sections, --170 lines):

- Remove 'Current Architecture Principles' -- owned by architecture
- Remove 'Portfolio Data Flow' -- owned by architecture
- Remove 'Portfolio Accounting Rules' -- owned by investment-domain
- Remove 'Source Of Truth' -- owned by investment-domain
- Remove 'Currency Rules' -- owned by investment-domain

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- Remove 'Market Data Architecture' -- owned by architecture
- Remove 'Safe Zones (brief)' -- owned by safe-zones
- Remove 'Current Governance Rule' -- owned by architecture (Review First section)
- Remove 'Test Governance' -- owned by test-protocol
- KEEP: Purpose, Mandatory Loading Rule, Project Name, Product Vision, Current Dev Stage, Long-Term Goal
- Result: project-state shrinks from 238 to ~65 lines with no loss of unique information

From bousala-investment-domain (remove 1 section):

- Remove 'Research Separation' -- architecture owns 'Research Rules'. Keep one, delete the other.

7. Risks If We Do Nothing

Probability	Risk	Consequence
High over time	Rule updated in one skill but not duplicates	Agent loads contradictory rules; behaviour becomes unpredictable
Exists NOW	SAHMK 'Planned' in project-state	Agent may skip SAHMK safe-zone protection entirely
Exists NOW	CASH absent from architecture test list	Agent skips cash regression coverage
Exists NOW	Review First format diverges	Inconsistent review output across sessions
High	Session state gap G1 unaddressed	alloc_ms_market class of conflict repeats for every new filter pair
Med	dev_test_runner.py unprotected	Test count can drop silently; governance broken at foundation

8. Risks If We Consolidate Now

Probability	Risk	Mitigation
Low-Med	Edit introduces an omission	A rule is lost in merge; mitigated by careful diff review before each change
Low	Frontmatter trigger breaks	Agent stops loading a skill; mitigated by updating descriptions at the same time
Low	project-state becomes too thin	Loading sequence loses authority; keep it as the primary purpose of the file
Med	One session consumed	One-time investment; not a recurring cost

9. Recommended Maximum Governance Footprint

- Maximum 5 skills at current MVP stage
- Maximum 150 lines per skill after consolidation
- Maximum 700 lines total loaded context per session
- Current state: 5 skills, 1,077 lines -- 35% is duplication
- After consolidation: estimated 5 skills, 650-680 lines -- within ceiling
- Consolidate content inside existing skills. Do NOT reduce the skill count.

10. Recommendation: Consolidate Some Skills

The five-skill structure is correct and the boundary definitions are sound. The problem is that content migrated into new skills was not removed from old ones, so duplication accumulated. Recommended action: targeted removal of duplicated sections only -- not a full rewrite. Estimated effort: one focused session.

Skill Classifications:

Skill	Classification	Rationale
bousala-safe-zones	Essential -- Highest ROI	Unique (P0-P4, escalation, minimal-change). Low cost. Loaded every implementation.
bousala-test-protocol	Essential -- High ROI	Unique (runner rules, failure/completion rules). Low cost. Every implementation.
bousala-investment-domain	Essential -- High ROI	Unique domain model. Low cost. Critical for accounting correctness.

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bousala-architecture	Valuable -- Needs trimming	Owns Review First (high value) but carries 4 sections duplicating dedicated skills.
bousala-project-state	Valuable -- Needs significant trimming	80% is restatement. Unique value = loading sequence + 3 items only.

Prioritised action list (do not implement until explicitly confirmed):

Priority	Effort	Action
P1	Trivial	Add CASH to architecture test category list (one line)
P2	Trivial	Correct SAHMK status in project-state from Planned to Implemented
P3	Low	Add session state ownership rule to safe-zones (closes G1 -- highest-likelihood future defect)
P4	Trivial	Add dev_test_runner.py to safe-zones protected list (closes G2)
P5	1 session	Strip project-state of 9 restated sections -- reduce from 238 to ~65 lines
P6	Low	Remove Safe Zones and Testing Rules sections from architecture (owned by dedicated skills)
P7	Trivial	Remove Single Source Of Truth from architecture -- investment-domain owns it
P8	Trivial	Unify Review First fields: add recommendation + architecture impact to architecture checklist
P9	Low	Add post-implementation app validation checklist to test-protocol (closes G4)

End of report. No files were modified. All recommendations require explicit approval before implementation.